

CE 501: Remote Sensing

Course Code	CE 501
Course Name	Remote Sensing
L-T-P-C	2-0-2-3
Prerequisites	–
Intended for	UG 4th year/PG
Distribution	Elective
Approval	8th Senate (Oct 2015)

Course Contents

- **Module I**
Introduction, principles and types of remote sensing Energy source and radiation principle.
Interaction of EMR (electromagnetic radiation) with atmosphere- absorption, scattering, Atmospheric windows, Interaction of EMR with Earth Surface-Spectral reflectance curves[5 Lectures]
- **Module II**
Platforms and Sensors: Geostationary and sun-synchronous orbits, Active and passive sensors, Spectral, spatial, temporal and radiometric resolutions.
Salient features of LANDSAT, SPOT, IRS satellites. [5 Lectures]
- **Module III**
Elements of Image interpretation, Visual analysis of data in VNIR (Visible and Infra-red).
Image formats, data histogram and image info. [4 Lectures]
- **Module IV**
Digital Image processing: atmospheric, radiometric & geometric corrections.
Image enhancement, contrast stretching-linear and non-linear, filtering, image ratios or indices, image transformations- PCA (principal component analysis) [6 Lectures]
- **Module V**
Classification: Supervised and Unsupervised Classification
Accuracy assessment and Kappa statistics [5 Lectures]
- **Module VI**
Application of digital image processing to various problems. [3 Lectures]

Text and Reference Books

1. J. R. Jensen, **Remote Sensing of the Environment an Earth Resource Perspective**, Pearson Education, 2003.
2. T.M. Lillesand and R.W. Kiefer, **Remote Sensing and Image Interpretation**, 6th Edition, John Wiley & Sons, 1994.
3. J.B. Campbell, **Introduction to Remote Sensing**, Taylor & Francis, London, 1996.
4. F.F. Sabins, **Remote Sensing: Principles and Interpretation**, W.H. Freeman and Company, 1997.
5. J. R. Jensen, **Introductory Digital Image Processing**, 3rd Edition, Prentice Hall, 2006.
6. G. Joseph, **Fundamentals of Remote Sensing**, Universities Press, 2003.
7. A.K. Keshari, **Satellite Remote Sensing**, Wiley, 2015.